

ENTITY RELATIONSHIP ANALYSIS REPORT

QuickBuy – E-Commerce Order Management Database System

1. Introduction

This report analyzes the relationships among entities in the QuickBuy E-Commerce Order Management Database System and defines their cardinalities.

2. Relationship Analysis

1. Customer and Orders

- One customer can place many orders.
- Each order belongs to only one customer.

Relationship: Customer → Orders

Cardinality: One-to-Many (1:M)

2. Category and Product

- One category contains many products.
- Each product belongs to only one category.

Relationship: Category → Product

Cardinality: One-to-Many (1:M)

3. Supplier and Product

- One supplier can supply many products.
- Each product is supplied by one supplier.

Relationship: Supplier → Product

Cardinality: One-to-Many (1:M)

4. Orders and Order_Details

- One order contains multiple order detail records.
- Each order detail belongs to one order.

Relationship: Orders → Order_Details

Cardinality: One-to-Many (1:M)

5. Product and Order_Details

- One product can appear in many orders.
- Each order detail refers to one product.

Relationship: Product → Order_Details

Cardinality: One-to-Many (1:M)

6. Orders and Payment

- Each order has one payment record.
- Each payment belongs to one order.

Relationship: Orders ↔ Payment

Cardinality: One-to-One (1:1)

7. Orders and Shipment

- Each order has one shipment record.
- Each shipment belongs to one order.

Relationship: Orders ↔ Shipment

Cardinality: One-to-One (1:1)

8. Customer and Review

- One customer can provide many reviews.
- Each review belongs to one customer.

Relationship: Customer → Review

Cardinality: One-to-Many (1:M)

9. Product and Review

- One product can receive many reviews.
- Each review belongs to one product.

Relationship: Product → Review

Cardinality: One-to-Many (1:M)

3. Many-to-Many Relationship

The relationship between Orders and Products is Many-to-Many because:

- One order can contain multiple products.
- One product can appear in multiple orders.

This relationship is resolved using the **Order_Details** entity.

Orders ← **Order_Details** → **Products**

Cardinality: Many-to-Many (M:N)

4. Cardinality Summary

Relationship	Cardinality
Customer - Orders	1 : M
Category - Product	1 : M
Supplier - Product	1 : M
Orders - Order_Details	1 : M
Product - Order_Details	1 : M
Orders - Payment	1 : 1
Orders - Shipment	1 : 1
Customer - Review	1 : M
Product - Review	1 : M
Orders - Products	M : N

5. Conclusion

The QuickBuy E-Commerce Order Management Database System consists of well-defined entities and relationships that support customer management, product management, order processing, payments, shipment tracking, and customer reviews while maintaining database integrity and reducing redundancy.